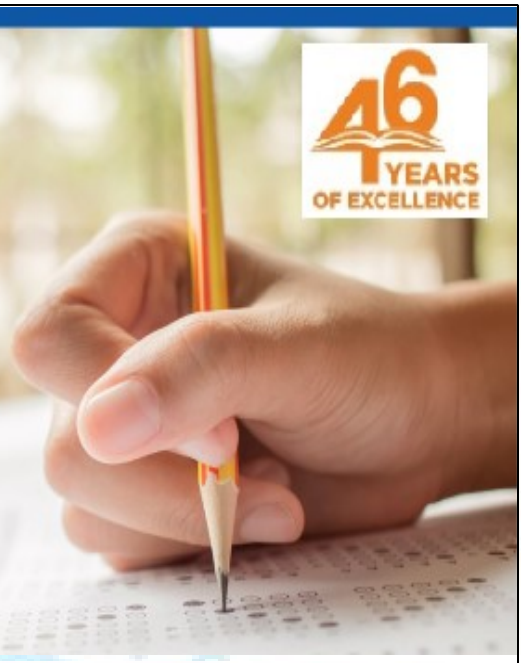
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 26-04-26
Time: 3 Hrs Max. Marks: 180

26-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-9(P1)_SYLLABUS

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MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 10)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 11 – 16)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 23)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 24 – 29)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 30 – 35)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 42)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 43 – 48)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 49 – 54)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. For non-negative integers n, define the function $h_n(x)$ as $h_n(x)=\sum_{k=1}^{n}\frac{\sin kx}{k+\cos kx}$, where $x\in\mathbb{R}-(2m+1)\pi, m\in\mathbb{Z}$. Let $h'_n(x)$ denote the derivative of $h_n(x)$.

Which of the following statement is INCORRECT?

A) There exists $\alpha\in(0,2\pi)$ such that $h'_n(\alpha)=0$, for all $n\geq 1$

B) $h_3\left(\frac{\pi}{2}\right)=\frac{2}{3}$

C) for all $n, h_n(x)$ is an odd function

D) $h_n(x)$ periodic with period $\frac{2\pi}{n+1}$

2. If the area of region $\mathbb{R}=\{(x,y)\in\mathbb{R}\times\mathbb{R}: 0\leq x\leq 2; 0\leq y\leq 1\}$ and $\frac{1}{3}\leq xy\leq 1$ is λ square unit, Then λ is equal to

A) $\frac{2}{3}+\frac{2}{3}\ln 2-\frac{1}{3}\ln 3$ B) $\frac{3}{2}+\frac{3}{2}\ln 2-\ln 3$ C) $\frac{2}{3}-\frac{2}{3}\ln 2+\frac{1}{3}\ln 3$ D) None of these

3. $^nC_0x^{2n}+\frac{^nC_1}{2}x^{2n-2}\left(2-x^2\right)+\frac{^nC_2}{3}x^{2n-4}\left(2-x^2\right)^2+....+\frac{^nC_n\left(2-x^2\right)^n}{(n+1)}=$

A) $\frac{2^n-x^{2n+2}}{(n+1)\left(2-x^2\right)}$ B) $\frac{2^n-x^{2n}}{(n+1)\left(2-x^2\right)}$ C) $\frac{2^{n+1}-x^{2n+2}}{(n+1)\left(2-x^2\right)}$ D) $\frac{2^{n+1}-x^{2n}}{(n+1)\left(2-x^2\right)}$

4. Let $a_1, a_2, a_3, \ldots\ldots\ldots, a_n$ be real numbers different from 1 and let n be a natural number such that

$a_1^2+a_2^2+a_3^2+\ldots\ldots+a_n^2=1000$ and $\frac{a_1^2}{1-a_1}+\frac{a_2^2}{1-a_2}+\frac{a_3^2}{1-a_3}+\ldots\ldots+\frac{a_n^2}{1-a_n}=100$.

Let $S=\frac{a_1^3}{a_1-1}+\frac{a_2^3}{a_2-1}+\frac{a_3^3}{a_3-1}+\ldots\ldots+\frac{a_n^3}{a_n-1}$. Then the number of ways in which S can be expressed as a product of two integers is

A) 15 B) 18 C) 28 D) 36

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SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 5 and 6:

Suppose $f(x)$ is a function satisfying the following conditions:

(i) $f(0)=\frac{1}{2}, f(3)=-\frac{13}{4}$;

(ii) $f(x)$ has a maximum value at $x=-1$;

$f'(x)=\begin{vmatrix} 2px & 2px-1 & 2px+q+1 \\ q+2 & q+1 & 1 \\ 2px+q+3 & 2px+q+1 & 2px+q+2 \end{vmatrix}$. If $p_1=f(1)$ and range of $f(x)$ as $(-\infty, p_2]$, then

5. p_1 is equal to

6. p_2 is equal to

Question Stem for Question Nos. 7 and 8:

$f, g: R \to R$ are functions such that

$f(x)=3x-\sin\left(\frac{\pi x}{2}\right), g(x)=x^3+2x-\sin\left(\frac{\pi x}{2}\right)$

7. The value of $\left(\frac{d}{dx}\left(f^{-1}\left(g^{-1}(x)\right)\right)\right)_{x=12}$ is equal to $\frac{2}{3k}$ then k is equal to

8. The area bounded by $y=f^{-1}(x), y=g^{-1}(x)$ is

Question Stem for Question Nos. 9 and 10:

Let α, β be real numbers such that the system of linear equations $x+y+z=2$ $x+2y+3z=5$ $x+3y+\alpha z=\beta$ has infinitely many solutions. Let $|M|$ represent the determinant of the matrix $M=\begin{bmatrix} -2 & -3+\beta & \sin\theta-2 \\ 1 & \sin\theta+2 & \gamma \\ \alpha & 2\sin\theta-1 & -\sin\theta+2+2\gamma \end{bmatrix}$ Where $\gamma \in R, \theta \in [0, 2\pi]$ and minimum value of determinant of M is 8 and foot of perpendicular from the point $P(\alpha-4, \beta-10, 1)$ to the plane $x+2y-2z-10=0$ is $F(a,b,c)$, then

9. The value of $\frac{\gamma}{5}$ is ____ .

10. The value of $\frac{3}{2}(a+b+c)$ is ____ .

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

11. Consider the equation $3x^7 - x^4 - 30x^5 + 10x^2 + 3x^3 - 1 = 0$ then

A) The minimum real root of equation is $\left(-\sqrt{3}-\sqrt{2}\right)$

B) The minimum real root of equation is $\left(-\sqrt{3}+\sqrt{2}\right)$

C) The maximum real root of equation is $\left(\sqrt{3}+\sqrt{2}\right)$

D) The number of positive roots of equation is 3

12. For all real values continuous, differential and strictly increasing function f on the real line for all x satisfy the condition $f^2(x) = \int_0^x \left[(f(t))^2 + \left(f'(t)\right)^2 \right] dt + 2025$. Then the area bounded by curve $y = \dfrac{f(x)}{\sqrt{2025}}$ and lines $x = 0$ and $y = e$ is/are

A) $\dfrac{3}{\pi^3}\int_{-\pi}^{\pi} \dfrac{x^2}{1+\sin x + \sqrt{1+\sin^2 x}}\,dx$

B) $e^{2025} - \int_0^1 e^x dx$

C) $\int_1^{e^{2025}} \ln y\,dy$

D) $\int_1^e \ln(e+1-y)\,dy$

13. Let $f:(0,2\pi) \to \mathbb{R}$ be a differentiable function such that $\lim\limits_{t\to x} \dfrac{f(x)\cos t - f(t)\cos x}{t-x} = \dfrac{\cos^2 x}{x^2}; \forall x \in (0,2\pi)$. If $f(\pi) = -\dfrac{1}{\pi}$, then which of the following statement(s) is/are TRUE ?

A) $f(x)$ has local minimum in the interval $\left(\dfrac{\pi}{2}, \pi\right)$

B) $f(x)$ is increasing for $x \in \left(0, \dfrac{\pi}{3}\right)$

C) $\int_{1/2}^1 x^3 f'(x)\,dx < 0$

D) $\dfrac{7}{30} \leq \int_0^1 x^3 f(x)\,dx < \dfrac{1}{3}$

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14. The integers from 1 to 1000 are written in order around a circle. Starting at 1, every fifteenth number is marked (i.e. $1,16,31,\ldots.$). This process is continued until a number is reached which has already been marked, then marked numbers are

A) 11 B) 17 C) 986 D) 996

15. Let a,b,c be distinct non zero complex numbers with $|a|=|b|=|c|$ and each of the equations $az^2+bz+c=0$ and $bz^2+cz+a=0$ has a root having modulus 1, then

A) $b^2=ac$ B) $c^2=ab$ C) $|a-b|=|b-c|=|c-a|$ D) $a^2=bc$

16. Consider the system of equations $ax+by=0$ and $cx+dy=0$ where $a,b,c,d\in\{1,2\}$. Then the probability that the system of equations has

A) unique solution is $\frac{1}{2}$ B) unique solution is $\frac{5}{8}$

C) non trivial solutions is $\frac{3}{8}$ D) non trivial solutions is $\frac{1}{2}$

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. If the coefficient of $x^{\frac{n^2+n-14}{2}}$ in $(x-1)(x^2-2)(x^3-3)(x^4-4)\ldots(x^n-n), n\geq 30$ is equal to k, then $\left[\frac{3k}{4}\right]$ is ? (where [.] is the greatest integer function)

18. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be given
$f(x)=|X^2-3X+2|+3$ and $g(x)=x^3+3x^2+3x+1$

define $h(x)$ as : $h(x)=\begin{cases}\max\{f(x),g(x)\}, x\leq 1\\ \min\{f(x),g(x)\}, x>1.\end{cases}$ If ℓ = number of point(s) where $h(x)$ is discontinuous and m = number of point(s) where $h(x)$ is not differentiable. Then find $\left|36\int_{\ell}^{m}|x^2-3x+2|dx\right|$

19. Number of integral values of x in the interval $[0,5\pi]$ which satisfy $\cos^9x-\sin^9x<5(\sin^5x-\cos^5x)$ is ____

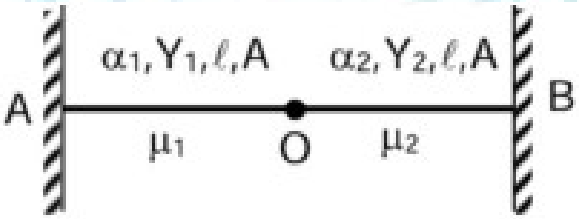
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PHYSICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

20. Two wires of same length $l = 1\text{ m}$ and same cross-sectional area $A = 10\text{ mm}^2$ are welded together and their ends are rigidly clamped between two walls as shown in the figure.



Given $\alpha_1 = 2\times10^{-5}\text{C}^{-1}, Y_1 = 1\times10^{10}\text{ N/m}^2, \mu_1 = 0.4\text{ kg/m}$

$\alpha_2 = 1\times10^{-5}\text{C}^{-1}, Y_2 = 2\times10^{10}\text{ N/m}^2, \mu_2 = 0.1\text{ kg/m}$ .The tension in the wires is produced by reducing the temperature of system by 20°C. Find the first overtone frequency (in Hz) of transverse vibration of the system with joint 'O' as a node.

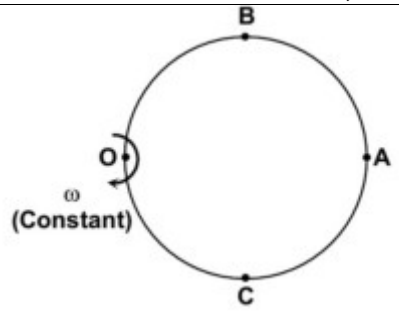
A) 10 B) 20 C) 30 D) 40

21. The pitch of a screw gauge is 1 mm and its circular scale is divided into 100 divisions. When nothing is placed between its studs, the zero of the circular scale lies 6 divisions below the reference line, and the zero of main scale is visible. When a wire is placed between its studs, the main scale reading is 2 mm and 56th division of circular scale coincides with the reference line of the main scale. Find the radius of the wire in cm .

A) 0.250 cm B) 0.125 cm C) 0.256 cm D) 0.128 cm

22. Suppose a uniform steel ring (mass M and radius R) is kept on a smooth horizontal surface with its plane horizontal. It is hinged about a point O on the perimeter & is made to rotate in a horizontal plane as shown (A, B, & C are points on the ring). The angular velocity ω is constant in time. The tension developed in the ring at point B is:

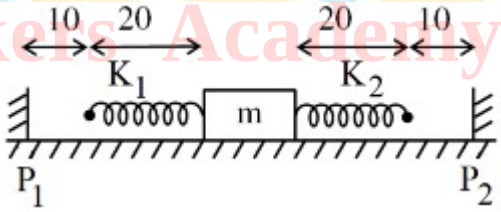
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A) $\frac{1}{2}M\omega^2R\left(\frac{2}{\pi}+1\right)$ B) $\frac{1}{4}M\omega^2R\left(\frac{2}{\pi}+1\right)$ C) $\frac{1}{2\pi}M\omega^2R$ D) $\frac{1}{4\pi}M\omega^2R$

23. Two springs, each of unstretched length 20 cm but having different spring constants $K_1 = 1000 N/m$ and $K_2 = 3000\ N/m$ are attached to two opposite faces of a block of mass $m = 100\ g$ kept on a smooth horizontal surface as shown in the figure. The outer ends of the two springs are now attached to two pins P_1 & P_2 whose locations are shown in the figure, as a result of this, the block acquires a new equilibrium position. The block has been displaced by small amount from its equilibrium position and released to perform simple harmonic motion, then

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A) new equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}$sec.

B) new equilibrium position is at 25 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}$sec.

C) new equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{200}$sec

D) new equilibrium position is at 30 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{200}$sec

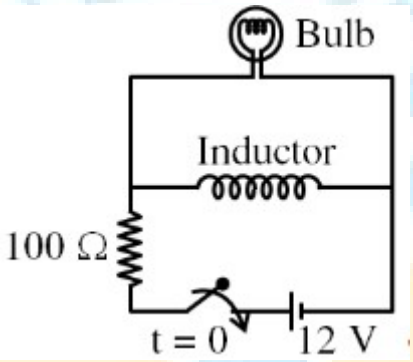
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SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 24 and 25:

A bulb (purely resistive of resistance 100Ω) is connected in a circuit using a DC battery of 12 V as shown:



The inductor has an inductance of 0.5 mH & resistance of 100Ω . Based on this, answer the following two questions

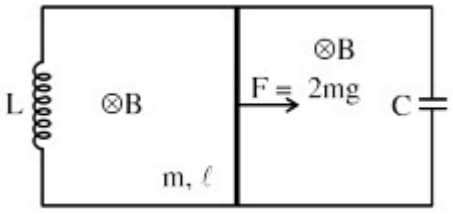
24. The magnitude of rate of change of current in the inductor, just after closing the switch measured in kA/s is ____ :

25. The bulb stops glowing if current passing through it becomes any less than 50 mA . Then for what time interval (in μs), after closing the switch, will the bulb glow for?

(Take $\ln 2 = 0.69$)

Question Stem for Question Nos. 26 and 27:

A conducting rod of mass ' m ' and length ' l ' is placed on a pair of smooth horizontal conducting parallel rails connected at one end by an inductor of inductance L and by a capacitor of capacitance ' C ' at the other end as shown in the figure. A uniform vertical magnetic field ' B ' exists in the region perpendicular to the plane of rails. The conductor starts from rest at $t = 0$ from the left end of the rails due to constant external horizontal force $F = 2mg$ applied to it. Considering $CB^2l^2 = m$, choose the CORRECT option(s):



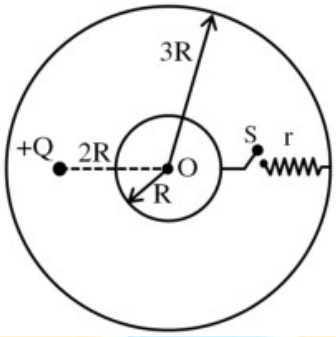
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26. The maximum horizontal displacement of the rod is $x\left(\frac{mgL}{B^2I^2}\right)$. Find the value of x.

27. The angular frequency of the oscillations of the rod is $\frac{BI}{\sqrt{y(\text{mL})}}$. Find the value of y.

Question Stem for Question Nos. 28 and 29:

Two uncharged thin concentric spherical conducting shells ' A ' and ' B ' of radii R and 3 R are connected through a resistance ' r ' and switch 'S' initially opened as shown in the figure.



A positive point charge ' Q ' is fixed at a distance $2R$ from the centre of the shells. At $t=0$, the switch ' S ' is closed till the steady state is reached. The charge on the outer surface of shell ' A ' after steady state reached is -xQ. The charge on the inner surface of shell ' B ' after steady state is reached is -yQ.

28. Find value of x.

29. Find value of y.

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

30. The magnitude of angular momentum, kinetic energy, velocity and frequency of an electron in a hydrogen atom corresponding to the quantum number n are L, K, v and f respectively. Then according to Bohr's theory of hydrogen atom:

A) $K^{-1}Lf$ is constant for all orbits.
B) Kvf^{-1} is constant for all orbits.
C) $K^{-1}Lv$ is proportional to n^2.
D) $K^{-1}f$ is proportional to $\frac{1}{n}$.

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31. A small sphere is charged uniformly and placed at some point $A(x_0, y_0)$ so that at point $B(9\text{ m}, 4\text{ m})$ electric field strength is $\vec{E} = (54\hat{i} + 72\hat{j})\text{NC}^{-1}$ and potential is 1800 V. Then :

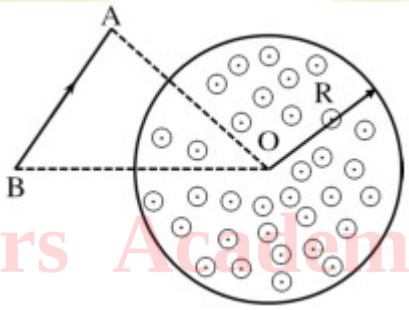
A) The magnitude of charge on the sphere is $4\mu C$

B) The magnitude of charge on the sphere is $2\mu C$

C) Coordinates of A are: $x_0 = -3, y_0 = -12$

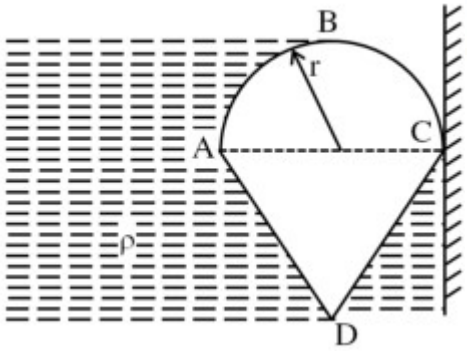
D) Coordinates of A are $x_0 = 4, y_0 = -1$

32. Figure shows cross-sectional view of a long solid cylinder of radius ' R '. Current ' I ' flows along outward direction along length of cylinder. Then, find possible value(s) of $\oint_{BA} \vec{B} \cdot \vec{d}_{along}$ BA . Given: $OB = (\sqrt{3}+1)^R, OA = 2R, \angle OBA = 45^\circ$. Where ' O ' is point of axis of cylinder.



A) $\frac{\mu_0 I}{6}$ B) $-\frac{\mu_0 I}{6}$ C) $\frac{\mu_0 I\sqrt{7}}{6}$ D) $\frac{-\mu_0 I}{12}$

33. Figure shows cross-sectional view of solid semi cylinder of radius ' r ' rigidly attached with solid triangular prism. It floats on a liquid having density ' p '. It is in contact with smooth vertical wall. Part BC is not in contact with liquid and the part AB is entirely inside the liquid.



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The length of objects is ' l '. Given ($AC = CD = DA$). The atmospheric pressure is P_0.

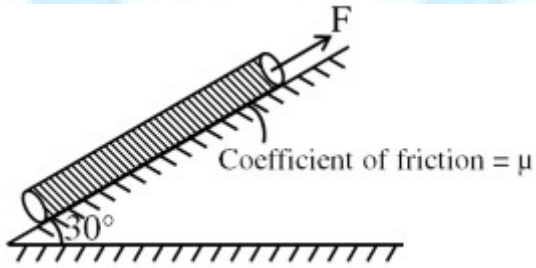
A) Buoyant force acting on object is independent of atmospheric pressure.

B) Normal force exerted by vertical wall is $\frac{pgr^2I}{2}$.

C) The force exerted by liquid on slant surfaces ADC of prism is $P_0(2rl)+\left(2+\sqrt{3}\right)r^2plg$.

D) Weight of object is $\left(\sqrt{3}+\frac{\pi}{2}\right)r^2lpg$.

34. A uniform wire of mass m , length L and cross-sectional area 'A', made of a material of Young's modulus Y , is pulled by a constant force F on an inclined plane such that it moves up the inclined plane as shown in figure. Choose the CORRECT opiotn (s).



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A) The extension in the wire is $\frac{Fl}{2YA}$

B) The tension in the rod at a distance $\frac{L}{4}$ from the lower end of the rod is $\frac{F}{4}$

C) The elongation potential energy (elastic) in the rod is

$\frac{1}{6}\frac{F^2\ L}{YA}$

D) The elongation potential energy (elastic) in the rod is $\frac{1}{4}\frac{F^2\ L}{YA}$

35. Potential energy of a system is given by $U(x)=(x+1)(x+2)$. Then:

A) Point $x=-\frac{3}{2}$ corresponds to equilibrium position of the system.

B) Point $x=-1$ and $x=-2$ corresponds to equilibrium position of the system.

C) System is in stable equilibrium position at $x=-\frac{3}{2}$

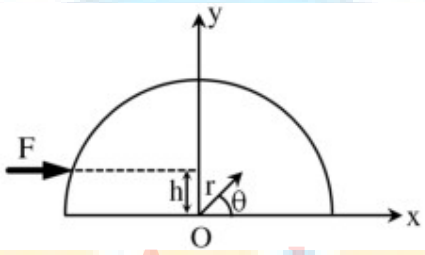
D) System is in unstable equilibrium position at $x=-\frac{3}{2}$

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SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

36. The surface mass density (mass per unit area) of a semi circular disc of radius ' R ' varies as $\sigma = |\vec{\alpha} \times \vec{r}|$, where α is a constant vector along positive x -axis and r is positive vector with respect to origin. It is kept on smooth horizontal surface with flat surface in contact with surface. Now, force 'F ' is applied in such a way that it moves translationally without any rotation. The value of h is $\frac{K\pi R}{16}$, then find value of k .



37. There is spherical symmetric charge distribution around origin. The electrostatic potential at a distance ' r ' from origin is given as $V = \frac{1}{4}e^{-Kr}$, where K is a positive constant. The charge present in a spherical volume of radius r is given as $(x)\pi r^{(y)} K\varepsilon_0 e^{-Kr}$. Find the value of $(x+y)$.

38. A uniform horizontal rod of length L and mass m is placed with its two ends on two supports A and B . The rod breaks suddenly at length $L/10$ measured from the support B. The reaction at support A immediately after the rod breaks is $\frac{x}{40}mg$. Find the value of x :

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CHEMISTRY **MAX.MARKS: 60**

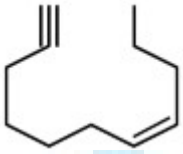
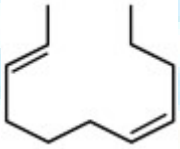
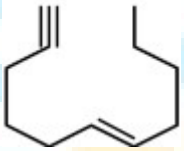
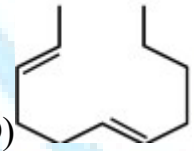
SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

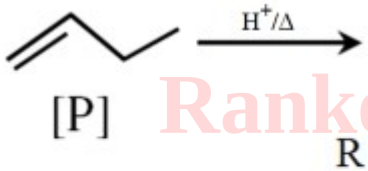
39. An acyclic diyne [P] with molecular formula $C_{11}H_{16}$ gives H_2 gas with sodium metal and also gives undecane on complete hydrogenation. Identify the major product in given reaction :

$$[P] \xrightarrow[\text{Naliq. } NH_3]{NaNH_2} [Q]$$

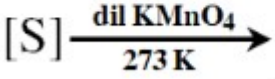
Major organic product

A)  B)  C)  D) 

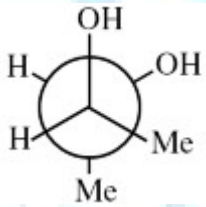
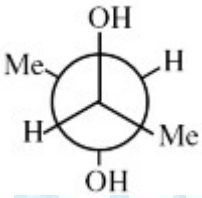
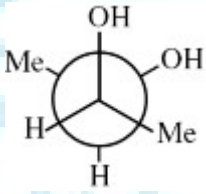
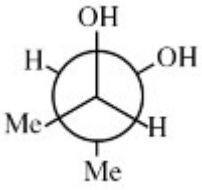
40. Identify the most stable conformer of [T] in the given sequence



R and S are isomers of [P]



Major product

A)  B)  C)  D) 

41. For an electrolytic cell undergoing following balanced reaction:

$$NO_3^- + 3H_3O^+ + Hg_2^{2+} \rightarrow 2Hg^{2+} + HNO_2 + 4H_2O$$

Under standard state conditions $E^\circ_{cell} = 0.02\ V$. If an average current of 10 amp . passes through the cell, how long will it take to produce 0.3 mole of HNO_2 by this reaction?

[Given : $NO_3^- + 3H_3O^+ + 2e^- \rightarrow HNO_2 + 4H_2O, E^\circ = 0.94\ V$ and

$\frac{2.303RT}{F} = 0.06,\ F = 96500 C mol^{-1}$]

A) 1.6 hours B) 16 hours C) 3.2 hours D) 32 hours

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42. At 298 K, E°_{OP} of the following reaction is

$$Ag(s)+2NH_3(aq)\rightarrow Ag(NH_3)^{+}{}_2(aq)+e^-$$

Given : $K_f\left(Ag(NH_3)_2{}^{+}\right)=10^{16}; E^{\circ}_{Ag^+/Ag=0.80\ V}$

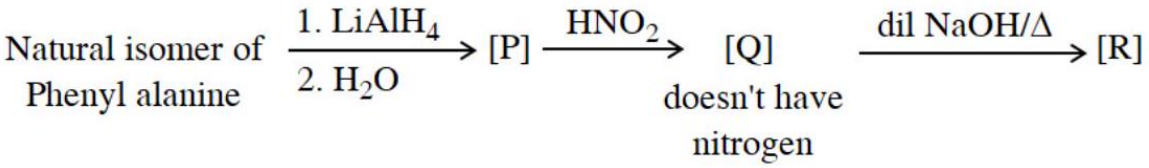
A) 0.144 V B) 1.744 V C) -0.144 V D) -1.744 V

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 43 and 44:

Consider the major organic product in each case:



43. Position of $-NH_2$ in [P] as per IUPAC is

44. Number of carbon atoms present in parent chain (as per IUPAC) in [R] is

Question Stem for Question Nos. 45 and 46:

Consider following reagents with given codes:

Code 1: dil. HCl Code 2: NH_3 (aq)

Code 3: KI (aq) Code 4: KCN (aq)

Code 5: $Na_2S_2O_3(aq)$ Code 6: $NaOH(aq)$

Let the sum of codes of reagents in which AgCl is soluble be x .

Let the sum of codes of reagents in which $Zn(OH)_2$ is soluble be y

45. Find $\frac{x}{2}$

46 Find $\frac{y}{4}$

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Question Stem for Question Nos. 47 and 48:

Cadmium oxide crystallises in NaCl type structure. The compound is however nonstoichiometric with the formula $CdO_{0.95}$. The defect arises due to some cationic positions occupied by neutral Cd atom in place of Cd^{2+} ions and equivalent numbers of anion sites are vacant.

$(Cd = 112,\ N_A = 6\times10^{23})$

47. What percent of anion sites are vacant?

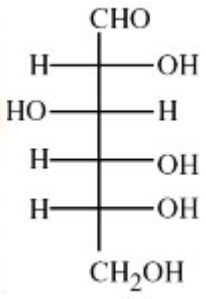
48. If the edge length of the unit cell is 400 pm , what would be the density (in gm/cm^3)]

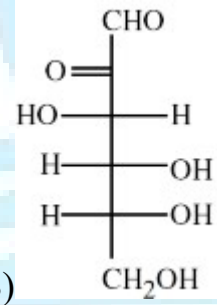
SECTION - 3 (Maximum Marks : 24)

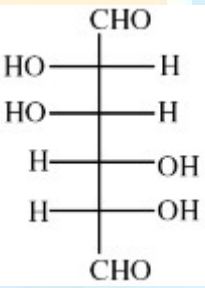
This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

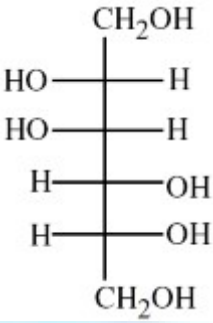
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

49. Treatment of D-glucose with aqueous NaOH results in a mixture of monosaccharides, identify the monosaccharide(s) present in the mixture :

A) 

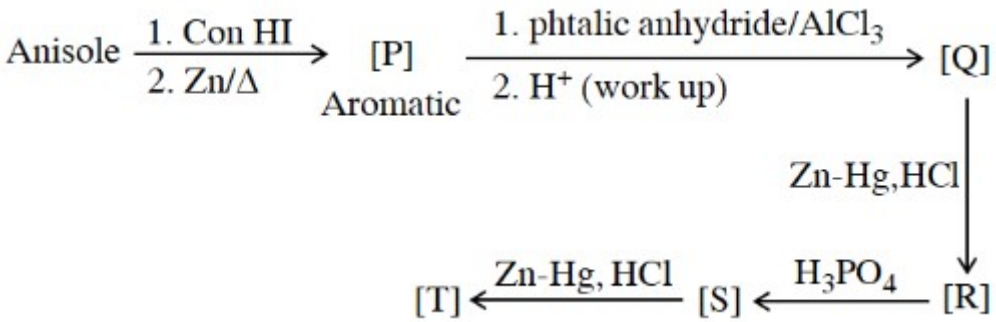
B) 

C) 

D) 

50. Considering the following reaction sequence, the CORRECT statement(s) is(are) (note in every

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stage consider major organic product

A) Compounds [Q] and [R] both give brisk effervescence with NaOH

B) Compounds [R] and [S] both react with hydroxyl amine giving corresponding oxime

C) Degree of unsaturation in [T] is 9

D) Compound [P] can give neutral $FeCl_3$ test

51. 0.5 moles of sodium chloride is dissolved in 150 moles of water. The solution is cooled to a temperature ' T ' K below the freezing point of solution so that some ice is formed. After separating the ice formed, the remaining solution is heated to normal boiling point of water, where the vapour pressure of solution is found to be 750 torr. Identify CORRECT option(s) on basis of this information (K_f for water $= 2\ K\ kg\ mol^{-1}$)

A) Mass of ice formed at ' T ' K is 1.350 kg

B) Temperature T is $\left(273-\frac{40}{27}\right)K$

C) Freezing point of original solution is $\left(273-\frac{20}{27}\right)K$

D) Relative lowering of vapour pressure for the final solution is $\frac{1}{76}$

52. Which of the following complex is/are optically inactive and as well as diamagnetic:

A) $\left[Co(NH_3)_4Cl_2\right]^{\oplus}$

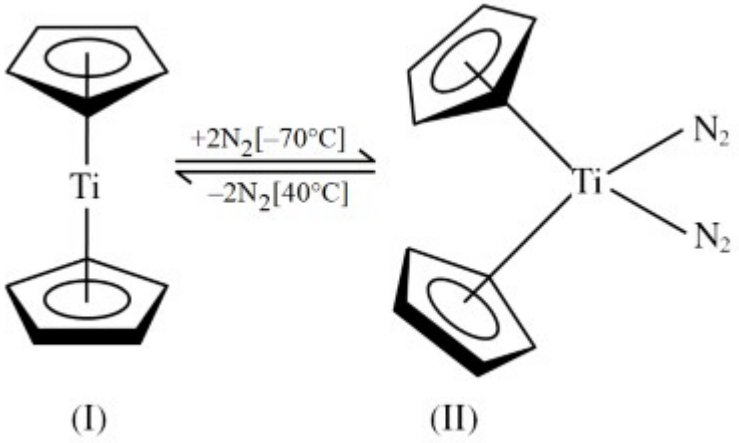
B) $\left[Cr(H_2O)_6\right]^{3+}$

C) cis $-\left[Co(NH_3)_2(en)_2\right]^{3+}$

D) $\left[PtCl_2(NH_3)_2\right]$

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53. The fixation of N_2 by certain bacteria is a biological reduction reaction of great interest. Very few nonbiochemical systems are known which will react with N_2 at room temperature and atmospheric pressure. One such compound is the titanium (II) compound titanocene, $(C_5H_5)_2Ti$ sometimes also abbreviated as Cp_2Ti, a sandwich compound like ferrocene. Select the CORRECT statement(s) about titanocene (I) & its dinitrogen complex (II) [Bond length of free N_2 109.8pm]



(I) (II)

A) The IUPAC name for the complex (II) would be bis(cyclopentadienyl) bis (dinitrogen) titanate (II)

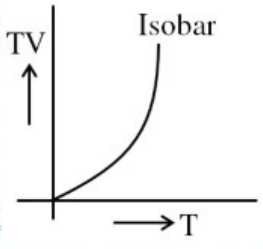
B) $N-N$ bond lengths in the complex will be greater than 109.8 pm

C) The $N-N$ bond length in the complex is less than 109.8 pm

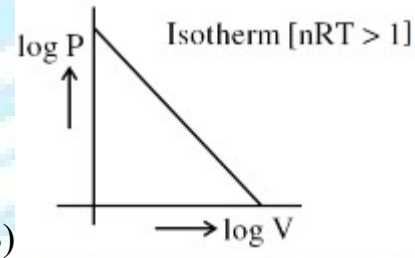
D) Of the two complexes, only complex (II) follows Sidgwick EAN rule

54. Which of the following plot(s) is/are CORRECT for a sample of ideal gas in given conditions ?

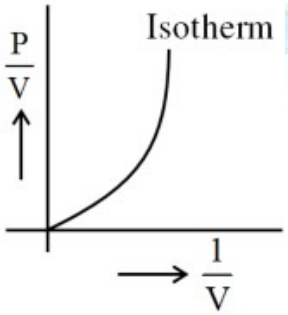
A)



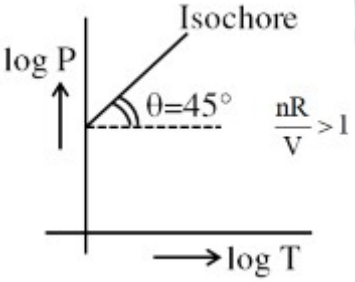
B)



C)



D)



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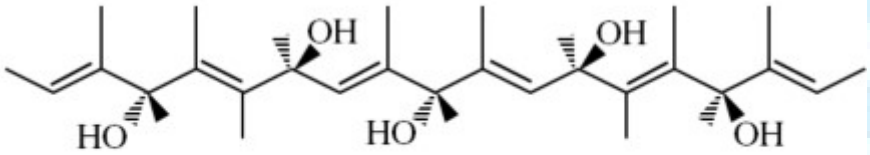
SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. The number of Sp^3 hybridized carbon atoms in [S]

C_5H_8 [P] liberate H_2 gas with Na $\xrightarrow[333K]{Hg^{2+}/H^+}$ [Q] [Q] gives 2-pentanol up on hydrogenation $\xrightarrow[-CHCl_3]{NaOCl}$ [R] $\xrightarrow[2.\ V_2O_5, 770K, 20\ atm]{1.\ NaOH,\ Electrolysis}$ [S]

56. The total number of different organic molecules formed from one molecule of P on complete ozonolysis ($O_3, Zn/H_2O$) is



[P]

57. How many of these reaction gives N_2 as product.

(i) $NH_4NO_3 \xrightarrow{\Delta}$

(ii) $NO \xrightarrow{\text{ed hot copper}}$

(iii) $Ba(N_3)_2 \xrightarrow{\text{heating}}$

(iv) $NH_4Cl + NaNO_2 \xrightarrow{\Delta}$

(v) $NH_2CONH_2 + HNO_2 \rightarrow$

(vi) $NaNO_3 \xrightarrow{\text{heating, } 800^\circ C}$

(vii) $NH_{3\ (excess)} + Cl_2 \rightarrow$

(viii) $(NH_4)_2Cr_2O_7 \xrightarrow{heating}$

(ix) $NH_4Cl + CaOCl_2 \rightarrow$

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